

From ImageNet to EuroSAT

Transfer Learning, When Transfer Helps, and Catastrophic Forgetting

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COMP6242 Deep Learning Group Project

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- 5-minute project video with contributions from all group members.
- Explain the problem, method, and findings clearly.
- Use visuals to support one coherent narrative rather than every detail.

Project Goal

- Main task: adapt an ImageNet-pretrained ResNet-18 to classify EuroSAT remote sensing scenes.
- Main question: does transfer learning help more than training from scratch?
- Supporting question 1: when does transfer help most?
- Supporting question 2: how much ImageNet capability is forgotten after adaptation?

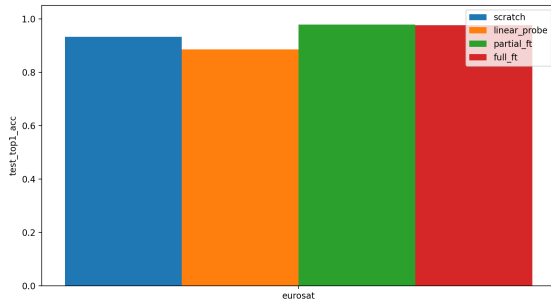
Experimental Setup

- Backbone: ResNet-18 with official ImageNet-1K pretrained weights.
- Main dataset: EuroSAT with 18,900 train / 4,050 val / 4,050 test images.
- Strategies: `scratch`, `linear_probe`, `partial_ft`, `full_ft`.
- Shared setting: 8 epochs, batch size 32, AdamW, learning rate 3×10^{-4} .
- Extra analyses:
 - Same-size cross-domain: EuroSAT vs CIFAR-10
 - Same-domain data-size: EuroSAT 10%, 30%, 60%, 100%

Main Result on EuroSAT

Strategy	Top-1	Macro-F1
scratch	93.21	93.04
linear_probe	88.54	88.33
partial_ft	97.78	97.71
full_ft	97.56	97.50

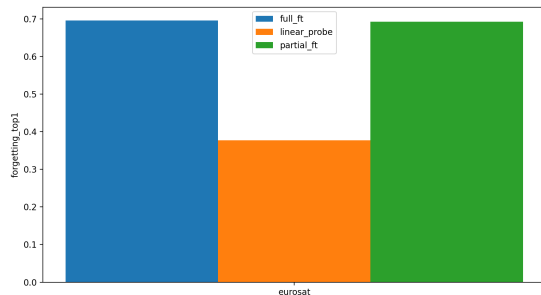
- Transfer helps only when the backbone is allowed to adapt.
- `partial_ft` gives the best EuroSAT performance.



Forgetting on ImageNet

Strategy	Forgetting	EuroSAT Top-1
linear_probe	37.73	88.54
partial_ft	69.26	97.78
full_ft	69.64	97.56

- Linear probing preserves more ImageNet ability.
- Stronger EuroSAT adaptation causes much larger forgetting.



When Transfer Helps

- Cross-domain comparison: CIFAR-10 receives larger transfer gain than EuroSAT.
- Data-size comparison: transfer is most valuable in the low-data EuroSAT setting.
- Together, these experiments show that transfer benefit depends on target domain and label budget.

Team Contributions and Takeaways

- Ruiquan Qiao: main experiment and corresponding forgetting analysis.
- Tiancheng Xia: same-size cross-domain analysis and corresponding forgetting analysis.
- Guangde Shi: data-size analysis and corresponding forgetting analysis.
- Takeaway: ImageNet transfer helps EuroSAT, but strong fine-tuning also causes severe forgetting on ImageNet.